

Sri Hari Sirisipalli

Lead Engineer & AI Systems Engineer · End-to-End

SUMMARY

Lead Engineer and AI Systems Engineer. I *lead, engineer, and build* end-to-end.

AI, computer vision, GenAI, apps, and industry software from concept to production. Digital twins, CFD, FEA, and reliability where the work needs deep physics. Cloud, DevOps, and product delivery to ship it. Marine, offshore, and naval engineering when the project sits in that domain. Fractional CTO at Openct.

The bet the practice is built on: engineering depth and product delivery are more useful together than apart. No handoff between the person doing the analysis and the person doing the app. Same engineer, both halves.

EXPERIENCE

Fractional CTO

Feb 2026 – Present · Remote

Openct

- Embedded as fractional CTO leading product audit, feature development, full openct.co website redesign, backend rebuild, and Flutter mobile app rebuild.
- New backend architecture from scratch on **DigitalOcean, Redis, PostgreSQL**. Third-party integrations across payments, communication, and analytics.
- CI/CD for mobile and backend. Observability, incident response, deployment automation.
- Owning architecture decisions, hiring input, code review, engineering standards, and product-engineering delivery cadence.

Lead Engineer & AI Systems Engineer

Jun 2023 – Present · Freelance · Remote

Independent Systems Engineering

Independent engineering practice building end-to-end systems across AI, digital twins, apps, backend, cloud, computer vision, and core engineering. Also leading engineering teams on marine, offshore, and naval design projects.

AI, ML & GENAI

- RAG systems, embeddings, semantic retrieval, and offline LLM stacks.
- **Real-time offline 3D AI avatar** (deployed): 75% STT and 52% TTS latency reduction. Whisper, Ollama, LangChain, WebSockets, Three.js.
- **DocForge**: offline document intelligence for engineering and compliance teams.
- **Ship Recycling Expert System**: HKC, Basel Convention, EU SRR, ILO compliance; 204-activity workflow DAG; 7 PDF report types.

COMPUTER VISION

- Multi-view cattle biometric retrieval at scale: 100,000+ images, SimCLR contrastive embeddings, FAISS multi-index, 20 distributed workers.
- Building end-to-end computer vision for **SCCL Mining Operations** (upcoming): 11 detection features, live NOC dashboard.

DIGITAL TWINS & SIMULATION

- Offshore riser fatigue and motion digital twin. Predicted significant wave height at R^2 0.9992 across 88,560 simulated sea states and 40,000+ ML experiments.
- Mooring digital twin, Battery digital twin, Ship emergency-response digital twin.
- MOSES sea-transport DOE automation platform: stratified LHS over 4 ballast zones, 15,000 DOE cases from 80 parameters. 12-sheet per-case Excel workbooks (6-DOF motions, structural loads, intact and damage stability across 28 compartments).
- **NavalArch Studio**: ship design and stability suite. 13 domain subpackages, ~450 tests, Maxsurf-validated (4,759 t displacement match).
- **PyNAOS**: pure-Python FEA engine for offshore structures. 593 tests at 98.6% coverage, six analysis drivers, custom Skyline LDLT solver.

RELIABILITY & ML FOR ENGINEERING R&D

- **Reliability Analysis Workbench** for underwater vehicle propulsion. OpenTURNS, DOE, ML surrogates over CFD and FEA.
- Multi-input DC-DC converter surrogate and inverse design: physics-informed network with 5-layer OOD gate, $\sim 10^4\times$ inference speedup.
- Vibration diagnostics rule engine (ISO 10816): 13 diagnostic rules, 30+ features, PyQt5 workflow.

APPS, PLATFORMS & BACKEND

- **zapstays.in** (live production): full-stack hostel and co-living platform. ~9,000 LOC PHP 8, 50 pages, 60 API endpoints, 20+ DB tables, real-time chat, Google OAuth, admin power-tools.
- **VHop** nightlife platform (in build).
- AWS (Lambda, EC2, S3, SQS, IAM), DigitalOcean, Redis, PostgreSQL. Terraform, Docker, CI/CD. Async and multiprocessing pipelines for high-throughput workloads.

MARINE, OFFSHORE, NAVAL & AUV

- Leading engineering teams on ship, offshore, and naval design projects. Hydrostatics, stability, seakeeping, damage. Offshore platforms, jack-ups, risers, mooring, gangways. Planning craft, service boats.
- Leading an engineering team on **Autonomous Underwater Vehicle (AUV)** design and development. Sub-100 m class, CFRP pressure hull, vectored thruster, autonomous survey missions. Requirements-driven MBSE and V-model with FMEA, concept through vehicle delivery.
- Class work against IMO IS Code, SOLAS, IACS, DNV-GL.

Software Engineer, LLM & Cloud Infrastructure

Mar 2024 – Present · Part-time · Remote

Pangeon

- Patent similarity and prior-art LLM system at production scale for **Intellipat**, one of Pangeon's portfolio startups. 1,000+ daily similarity evaluations.
- Embedding-driven retrieval integrated with LLM semantic scoring for automated prior-art ranking.
- Re-architected compute workloads from EC2 to AWS Lambda. Infrastructure costs down 30%, batch completion speed up 5–7 $\times$.
- Reproducible AWS provisioning with Terraform across accounts. Custom Terraform GUI for AWS resource management.

AI / ML Advisor

Mar 2025 – Aug 2025 · Remote

Agritech Startup

- Benchmarked commercial (OpenAI, Anthropic) and open-source LLMs (LLaMA, Mistral) for agriculture QA.
- Evaluated bilingual answering (English + Telugu) for Andhra agriculture use cases.
- Designed STT fine-tuning pipeline: audio collection, transcription, diarization, dataset prep.

Data Engineer

Mar 2023 – Jun 2023 · Full-time

Sas2Py

- Led migration of legacy SAS data pipelines to PySpark; over 99% functional parity on validation.
- Modeled job dependency graphs to eliminate scheduling bottlenecks and optimize distributed execution.

Machine Learning Intern

Jul 2022 – Dec 2022 · Remote

Corteva Agriscience

- Converted TensorFlow and PyTorch models to ONNX for cross-platform deployment.
- Optimized inference via structured pruning and post-training quantization; validated numerical consistency across formats.

Software Engineer Intern

Jun 2022 – Dec 2022 · Bangalore · Remote

Dojima Networks

- Integrated Polkadot ecosystem components for cross-chain interoperability.
- Implemented cross-chain communication for transaction routing and state sync. Deployed Prometheus + Grafana monitoring stack.

SELECTED PROJECTS

Openct — Product & Platform Rebuild

Fractional CTO · openct.co

Complete website redesign, Flutter mobile rebuild, backend on DigitalOcean + Redis + PostgreSQL, third-party integrations, CI/CD, and observability.

Offshore Riser Digital Twin

Motion & Fatigue

Coupled hydrodynamics (Bentley MOSES), FEA (custom Python engine + ANSYS APDL), and reliability. Live twin ingests measured motion and current data, updates fatigue budget in real time, flags risk components for operations.

Reliability Analysis Workbench

AUV Propulsion & Offshore Systems

DOE across CFD and FEA runs, ML surrogates, OpenTURNS reliability layer. Live dashboard for design-tweak to failure-probability feedback without waiting for a full solve.

LLM Patent Similarity Search

Pangeon / Intellipat

Embedding + LLM semantic scoring, EC2 to AWS Lambda re-architecture, Terraform GUI. 1,000+ daily evaluations at production scale.

zapstays.in — Hostel & Co-Living Platform

Live Production

~9K LOC PHP 8, 50 pages, 60 API endpoints, 20+ DB tables, real-time chat, Google OAuth, admin tools, CSV export, audit log.

Autonomous Underwater Vehicle (AUV)

Engineering Lead · Design & Development

Sub-100 m class body-of-revolution vehicle. CFRP pressure hull, vectored thruster, autonomous survey missions. Requirements-driven MBSE and V-model with FMEA, concept through vehicle delivery.

DocForge

Offline Document Intelligence

Local LLMs and embeddings, no internet round-trip. RAG over technical PDFs, patent corpora, and long specs. Python, LangChain, ChromaDB, React front end.

Real-Time Offline 3D AI Avatar

Deployed

Fully offline 3D agent. 75% STT and 52% TTS latency reduction. Whisper + Ollama + LangChain + WebSockets + Three.js.

Goodhar — Cattle Biometric ID

State Agritech Initiative

Large-scale multi-view cattle biometric identification. SimCLR embeddings, FAISS multi-index, near-real-time retrieval. On hold pending data collection.

PyNAOS & NavalArch Studio

Industry Software

PyNAOS: pure-Python FEA engine for offshore structures, 6 analysis drivers, custom Skyline LDLT solver, 593 tests at 98.63% coverage. NavalArch Studio: ship design + stability suite, 13 subpackages, Maxsurf-validated.

MOSES Sea-Transport DOE Automation

Simulation Platform

15,000 DOE cases over 80 parameters. PyQt5 orchestrator. XGBoost / PyTorch / LightGBM surrogates. SALib sensitivity. Optuna / PyMoo optimization.

DC-DC Converter Surrogate

Physics-Informed ML

Physics-informed network with 5-layer OOD gate. XGBoost R^2 0.985. $\sim 10^4\times$ inference speedup. PyQt5 GUI for parameter pinning.

SKILLS

AI, ML & GenAI

PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM. Hugging Face, Transformers, ONNX. Surrogate Modeling, Physics-Informed NNs, OOD Detection, SHAP, Feature Engineering, Model Optimization.

GenAI & LLMs

LangChain, LlamaIndex, RAG Pipelines, FAISS, ChromaDB. Vector DBs, Embedding Models. Ollama, vLLM, Whisper STT, Prompt Engineering, Offline LLM Stacks.

Computer Vision

OpenCV, YOLO, BoT-SORT, SimCLR, ResNet. Tesseract, MediaPipe, Visual Similarity Search, Multi-View Embeddings, Ranking Fusion.

Digital Twins & Simulation

Digital Twins, CFD, FEA, Reliability (OpenTURNS). DOE, Latin Hypercube, SALib, Optuna, PyMoo.

Engineering CAD / CAE

ANSYS Fluent, ANSYS APDL, ANSYS Mechanical, OpenFOAM. Bentley MOSES, Autodesk Inventor, SolidWorks, FreeCAD, Fusion 360, AutoCAD, Maxsurf, Rhino.

Apps, Mobile & Web

Flutter, Dart, React, React Native, Next.js. TypeScript, Vite, Tailwind, PyQt5, PyQt6, Streamlit, HTMX.

Backend & Cloud

FastAPI, Django, Node.js, AWS (Lambda, EC2, S3, SQS, IAM), DigitalOcean. GCP, Redis, PostgreSQL, MySQL, Terraform, Docker, CI/CD, Nginx.

Data & Observability

PySpark, NumPy, SciPy, Pandas, Polars, Airflow. Prometheus, Grafana, MLflow, Weights & Biases, ONNX, ETL Pipelines.

Programming

Python, TypeScript, JavaScript, Dart, C++. MATLAB, SQL, Bash, Java, Go, C.

Marine, Offshore & Naval

Naval Architecture, Ship Design, Offshore Structures, AUV, Hydrodynamics. Seakeeping, Mooring, Stability. IMO IS Code, SOLAS, IACS, DNV-GL.

EDUCATION & LEADERSHIP

B.Tech, Mechanical Engineering

Mahindra University, Hyderabad

Aug 2018 – Jun 2022 · CGPA 7.5

Coursework: Linear Algebra & Matrices, Probability & Statistics, Data Structures, Big Data Computing, Advanced Data Analytics, Time Series Forecasting, GPU Programming, Computer Aided Engineering Design.

- **CODEIAM Hackathon, Andhra University** · Mentor (Jul 2024) · Mentored 3 student teams on AI, ML, and engineering software projects; 2 teams in top 5 of 40. Guided teams on problem framing, technical architecture, model selection, and demo delivery.
- **Zenith Science Club** (2019 – 2022) · Core team for MU Research Symposium 2020.
- **AERO Sports Meet** (2019 – 2022) · Volunteer and Security team for annual campus events.

PUBLICATIONS

ADMOS: An Adaptive Dual-Mode Operational Swarm for Resilient Asset Defense under Communication Jamming

Research Square · Preprint, under review at *Discover Vehicles* (Springer Nature) · Jul 15, 2026 · DOI: [gist.science/paper/rs/rs-9957095](https://doi.org/10.21203/rs.3.rs-9957095)

Mohamed Hijazy Shazin Hassan, Dandu S. N. V. P. R. Varma, Jaswanth Kumar Bobbadi, Neeraj Sree Mailee Dudaboyina, Narendra Sirisipalli, **Sri Hari Sirisipalli**, Giri Rajasekhar Gunnu.

LANGUAGES English (Fluent) · Telugu (Native) · Hindi (Conversational)